Biospectra : ISSN: 0973-7057, Vol. 20(2), September, 2025, pp. 233-240 **Plant Science**
An International Biannual Refereed Journal of Life Sciences

ISSN : 0973-7057

Studies on physico-chemical values of water from Kankrej taluka in Banaskantha District, North Gujarat

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Received : 24th June, 2025 ; Revised : 25th July, 2025
DOI:-https://doi.org/10.5281/zenodo.19198722

Abstract- In the present study water samples were collected from the Kankrej taluka available (three season) water in the lake was selected from seven villages. The present study is focused on the determination of physico-chemical parameters, such as Temperature, pH, EC, TDS, total hardness, calcium hardness, magnesium hardness, chlorides, free CO_2, DO, BOD, and COD of water samples from different lakes. The increase in pollution concentration indicates a significant rise in the pollution levels load due to domestic sewage and anthropogenic activities and the release of wastes into a lake at Kankrej taluka. The results revealed that the average pH value was analyzed as 7.38, Temperature was 30.36°C, Electrical Conductivity was 3.42 µS/cm, parameters include Total Dissolved Solids 502.29 mg/l, total hardness was 171.05 mg/l, calcium hardness 68.90 mg/l, magnesium hardness 24.81 mg/l, chloride was 367.51 mg/l, free CO_2 14.53 mg/l, dissolved oxygen was 1.38 mg/l, BOD was 0.88 mg/l, and COD was 125.71 mg/l of the lake water sample. Therefore, the study revealed that how the Lake water is contaminated by effluents from human activity and the dumping of waste from markets and domestic use wastages.

Keywords: Lake water, physico-chemical, Domestic waste, anthropogenic activities

INTRODUCTION

Fresh water reservoirs are one of the important water resources used for irrigation. Water is not only major component of environment but also universal solvent and a medium on which all organisms depend for their existence. Polluted water creates problems for all kind of life. Only 0.009% of total available fresh water is in lakes.[1] Water is the prime necessity of life, without it, there would be no life. Most of the biological reactions use water as the medium. So, the study of water bodies is equivalent to the study of life. Water is the habitat for a larger number of aquatic organisms ranging from microscopic plankton to large aquatic animals and macrophytes.[2]

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Water is essential in every stage of life for any living organism. It is the single most vital component of the earth that made it possible for life to originate, evolve, flourish and reach the present form that we have today. The earth is called a wet planet as two-third of it is occupied by water. About 99.7 percent of water found on the earth is in the oceans and seas, while it is not available for human consumption.[3]

Water is an important resource in the all living organisms, it is 70% of the earth surface occupies by water used for human development like drinking, irrigation and culturing of fishes. Lake water is the main source of drinking water, irrigation Kankrej taluka. Disposal of Sewage, Industrial wastes and other human activities effects of some lake were highly polluted.[4]

The present study was aimed at analyzing some important characteristics of lake water considered here in for selected village in kankrej taluka. Physico-chemical parameters such as pH, temperature, E.C., TDS, chloride content, Hardness, DO, BOD, COD, free CO_2 were carried out.

MATERIALS & METHOD

Study area

The total geographical area of the Kankrej sub-division is 845.80 square km. Out of that, the agricultural area of Kankrej covers 724.54 square km, and the rest areas cover 121.26 square km. The agricultural unit has been selected for the concerned research project which has an area of 725 sq. km. approximately.

Kankrej is a Taluka located in the Banaskantha district of Gujarat. It is one of 14 Talukas of Banaskantha district. There are 101 villages and 1 town in Kankrej Taluka. As per the Census India 2011, Kankrej Taluka has 48,979 households, a population of 2,75,613 of which 1,42,849 are males and 1,32,764 are females. The population of children between ages 0-6 is 44,364 which is 16.1% of the total population.

Banaskantha is one among the thirty-three districts of the Gujarat state of India. The administrative headquarters of the district is at Palanpur which is also its largest city. The district is situated between 23.33 to 24.45 north latitude and 72.15 to 73.87 east longitudes. The district is located in the Northeast of Gujarat and is presumably named after the West Banas River which runs through the valley between Mount Abu and Aravalli Range, flowing to the plains of Gujarat in this region and towards the Rann of Kutch**.** The district is bounded by Rajasthan in the north, Sabarkantha district in the east, Kutch in the west, and Patan and Mahesana district in the south.

Methodology

These studies carried out for seven different lakes from Kankrej taluka. There are 101 villages but selected village in Kankrej taluka. So that proper sampling method was necessary to get accurate result. The study was carried out from whole Kankrej taluka using key parameters. Studies on physico-chemical parameters are under taken with the objective of collection of water sample.

The water samples were collected in the polyethylene bottles or in a glass bottle. So initially the prewashed bottles were rinsed with the sample water then closed this bottle and dipped in the lake at 1ft, and then bottle was opened inside and was closed again to bring it out at the surface. The samples were collected from different points like surface water, centre point etc. Between the collection and analysis there were many reactions like physical, chemical and biological would be occurred so that sample analysis soon after the collection and sample preservation done by adding chemical preservatives, lowering the temperature and also by both methods.

This is done by standard methods. The temperature pH, dissolved oxygen, electrical conductivity and total dissolve solids were analysed immediately at a time of collection and other parameters were in the laboratory.

Table 1- Method of water samples analysis

Sl. No.	Characteristics	Methods used
1.	Temperature	Thermometer
2.	pH	pH Meter (Bio craft scientific systems)
3.	E.C	Conductivity method
4.	Ca^{+2}, Mg^{+2}	EDTA titrimetric method.
5.	Total Hard hardness	EDTA titrimetric method.
6.	TDS	Conductivity method
7.	Free CO_2	NaOH titrimetric method.
8.	Dissolved oxygen (DO)	Winkler method.
9.	BOD	5-day BOD test
10.	COD	Open reflux method.

RESULTS & DISCUSSION

The water samples were analysed for physicochemical characteristics. The physicochemical parameters were analysed namely Temperature, pH, EC, TDS, Total Hardness, calcium hardness, magnesium hardness, DO, COD, BOD, free CO_2, Chloride (Winter season Table 1 and summer season table 2).

Temperature: Temperature of water may not be as important in pure water because of the wide range of temperature tolerance in aquatic life, but in polluted water, temperature can have profound effects on dissolved oxygen (DO) and biological oxygen demand (BOD). The fluctuation in lake water temperature usually depends on the season, geographic location, sampling time and temperature of effluents entering the stream.[5] Water temperature was found to be winter season values of the samples ranged from 24.33°C to 32°C, highest in 32°C Rajpur Lake and lowest water temperatures were observed in Khoda Lake 24.33°C. Average temperature 29.28°C for selected lake. Water temperature was found to be summer season values of the samples ranged from 29°C to 39°C, highest in 39°C Khoda Lake and lowest water temperatures were observed in Zalmor and Mandla-1 Lake 29°C. Average temperature 31.4 °C for selected lakes. The variation is mainly related with the temperature of atmospheric and weather condition.[6]

PH: The pH values of the samples ranged from 7.21 to 8.01 in winter season, at highest pH values 8.01 Khoda Lake and lowest values 7.21 Rajpur Lake, Average pH 7.67 for selected lakes in winter season. PH values of the samples ranged from 6.40 to 8.63 in summer season, at highest pH values 8.63 Khoda Lake and lowest values 6.40 Gothda Lake, Average pH 7.09 for selected lakes in summer season. Ahmed and Rahman (2000)[7] reported that in most raw water sources pH lies in the range of 6.5- 8.5., where most of the water sample's different location tested in the study were found to be in the permissible range of pH value recommended by several health and pollution control organizations e.g., WHO, CPCB, BIS i.e., 6.5-8.5.

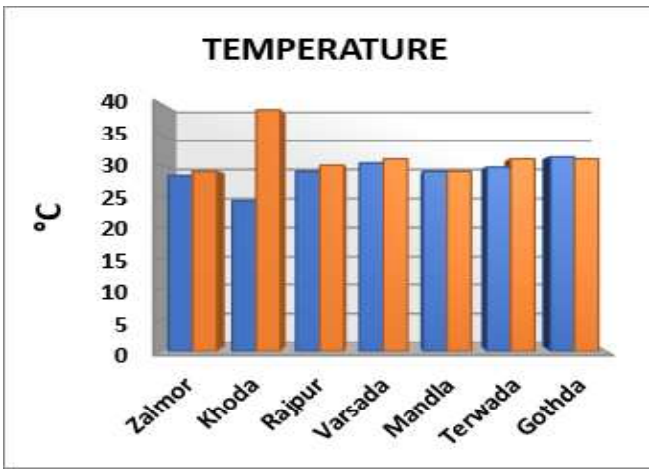
Electrical conductivity: Electrical conductivity value ranged from 0.658 to 9.51 µs/cm in winter season, at highest E.C. values 9.51 µs/cm Mandla-1 Lake and lowest values 0.658 µs/cm Gothda Lake, Average E.C. values 2.32 µs/cm for selected lakes in winter season. E.C. value ranged from 2.22 to 11.87 µs/cm in summer season, at highest E.C. values 11.87 µs/cm Rajpur Lake and lowest values 2.22 µs/cm Zalmor Lake, Average E.C. values 4.51 µs/cm for selected lakes in summer season. Usually, higher EC value indicate the presence of higher content of dissolved salts in lakes water[8] and the EC values are a good measure of the relative difference in water quality between different aquifers[9].

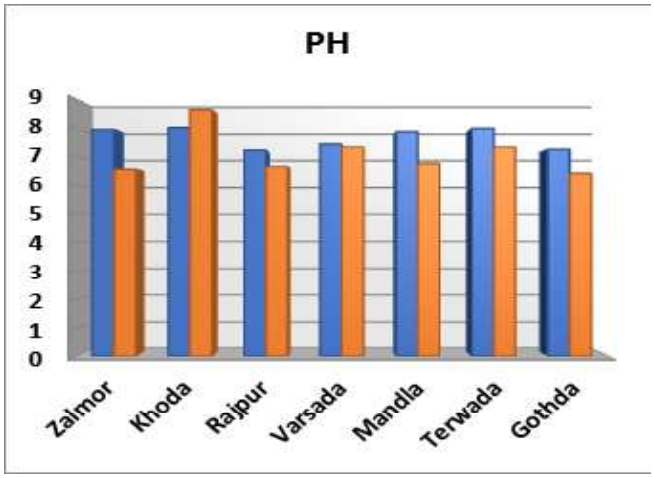
Table 1 - Winter Season Data

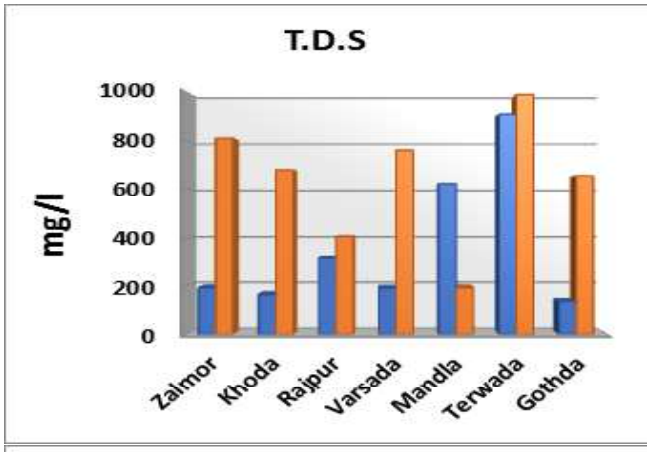
Sample name	Temperature (°C)	pH	E.C (µS/cm)	T.D.S (mg/l)	Total Hardness (mg/l)	Calcium Hardness (mg/l)	Magnesium Hardness (mg/l)	Chloride (mg/l)	Free CO_2 (mg/l)	DO (mg/l)	BOD (mg/l)	COD (mg/l)
Sample No.1 (Zalmor)	28.33	7.94	0.723	194	53.33	32.67	5.02	27.22	6.67	0.87	0.57	128
Sample No.2 (Khoda)	24.33	8.01	0.704	166	64.67	33.33	7.62	47.33	10	1.73	0.70	32
Sample No.3 (Rajpur)	29.00	7.21	1.225	319	50.67	27.33	5.67	131.35	11.67	0.57	0.83	32
Sample No.4 (Varsada)	30.33	7.43	0.735	192	42.67	30.00	3.08	130.17	6.67	0.97	0.80	224
Sample No.5 (Mandla-1)	29.00	7.86	9.51	623	167.33	48.00	29.00	868.57	11.67	1.97	1.20	64
Sample No.6 (Terwada)	29.67	7.98	2.68	909	53.33	28.00	6.16	42.6	20	0.93	0.93	288
Sample No.7 (Gothda)	31.33	7.24	0.658	138	32.00	25.33	1.62	63.9	23.33	1.37	0.23	64
MIN	24.33	7.21	0.658	138	32.00	25.33	1.62	27.22	6.67	0.57	0.23	32
MAX	32.00	8.01	9.51	909	167.33	48.00	29	868.57	23.33	1.97	1.2	288
AVE	29.28	7.67	2.32	363.00	66.29	32.09	8.31	187.31	12.86	1.20	0.75	118.86

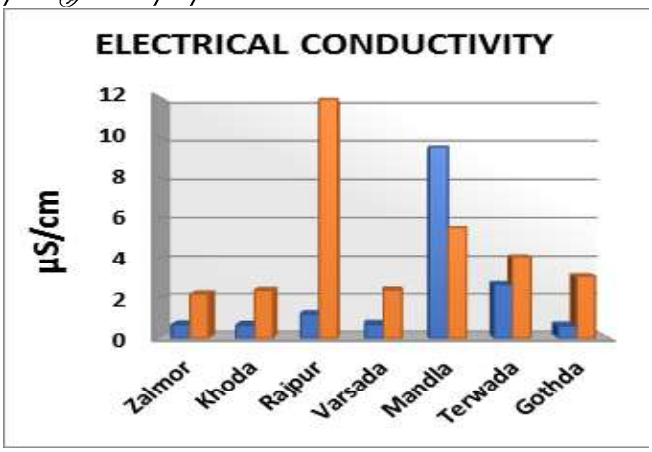
Table 2 - Summer Season Data

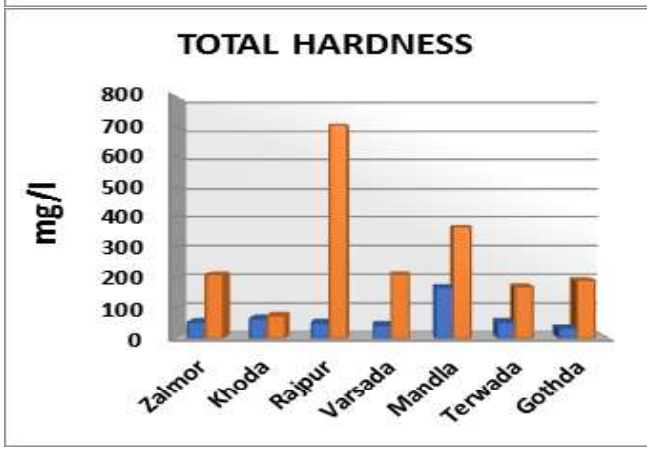
Sample name	Temperature (°C)	pH	E.C (µS/cm)	T.D.S (mg/l)	Total Hardness (mg/l)	Calcium Hardness (mg/l)	Magnesium Hardness (mg/l)	Chloride (mg/l)	Free CO_2 (mg/l)	DO (mg/l)	BOD (mg/l)	COD (mg/l)
Sample No.1 (Zalmor)	29.00	6.55	2.22	809	210	94	28.19	268.62	16.67	1.20	0.73	224
Sample No.2 (Khoda)	39.00	8.63	2.39	679	74.67	17.33	13.93	261.52	11.67	1.93	0.87	224
Sample No.3 (Rajpur)	30.00	6.64	11.87	406	706.67	208	121.18	1367.93	11.67	1.13	1.23	64
Sample No.4 (Varsada)	31.00	7.32	2.41	760	210.67	134.00	18.47	196.43	6.67	1.37	1.00	64
Sample No.5 (Mandla-1)	29.00	6.75	5.53	193	368	86.67	68.36	653.2	16.67	2.23	1.50	32
Sample No.6 (Terwada)	31.00	7.32	4.08	989	170	66.67	25.11	254.42	18.33	1.27	1.13	128
Sample No.7 (Gothda)	31.00	6.40	3.08	655	190.67	133.33	13.93	831.88	31.67	1.77	0.57	192
MIN	29.00	6.40	2.22	193	74.67	17.33	13.93	196.43	6.67	1.13	0.57	32
MAX	39.00	8.63	11.87	989.00	706.67	208.00	121.18	1367.93	31.67	2.23	1.50	224.00
AVE	31.43	7.09	4.51	641.57	275.81	105.71	41.31	547.71	16.19	1.56	1.00	132.57

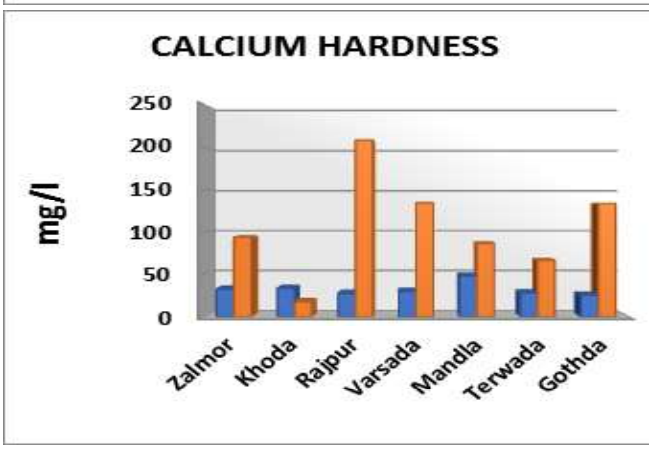


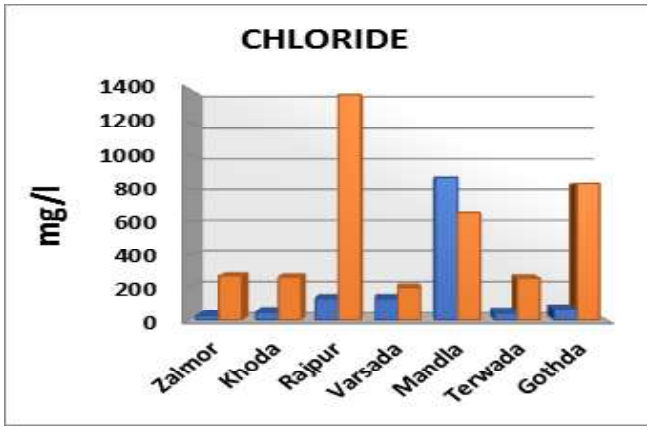


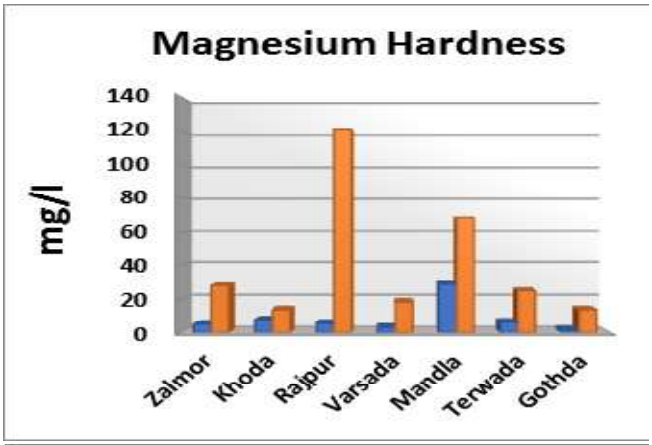

T.D.S
mg/l
1000
800
600
400
200
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda

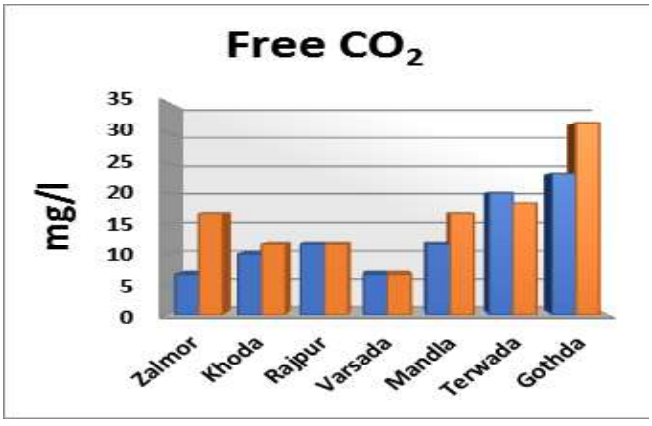

ELECTRICAL CONDUCTIVITY
µS/cm
12
10
8
6
4
2
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda

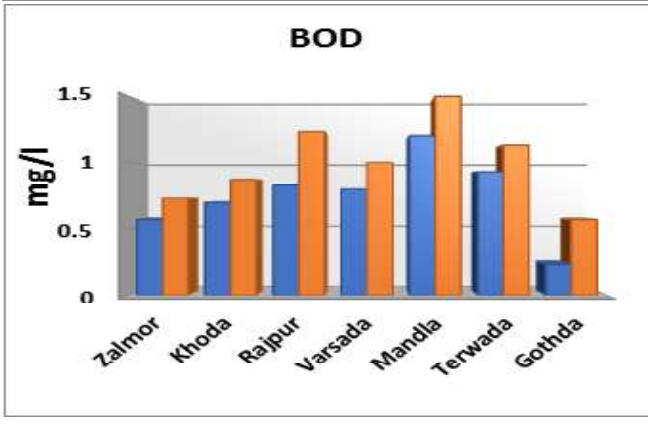

TOTAL HARDNESS
mg/l
800
700
600
500
400
300
200
100
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda

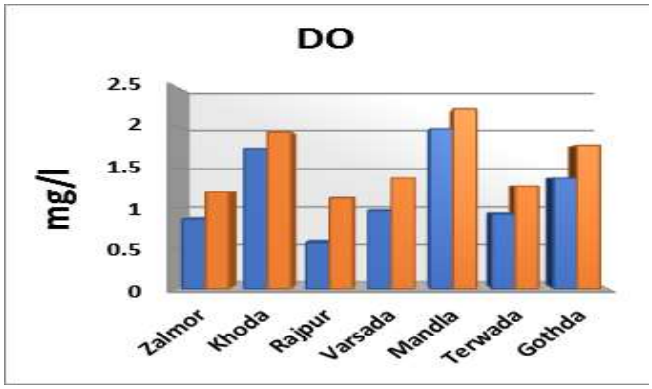

CALCIUM HARDNESS
mg/l
250
200
150
100
50
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda

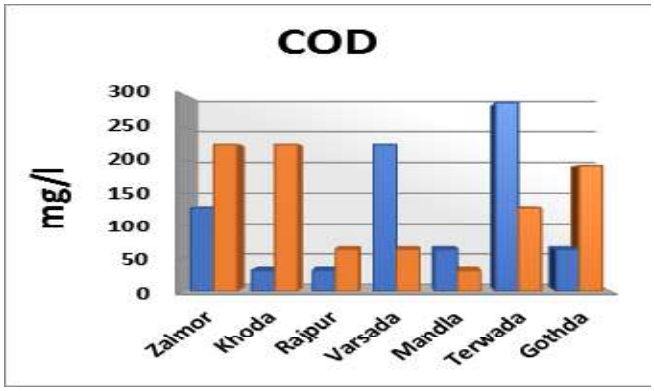

CHLORIDE
mg/l
1400
1200
1000
800
600
400
200
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda

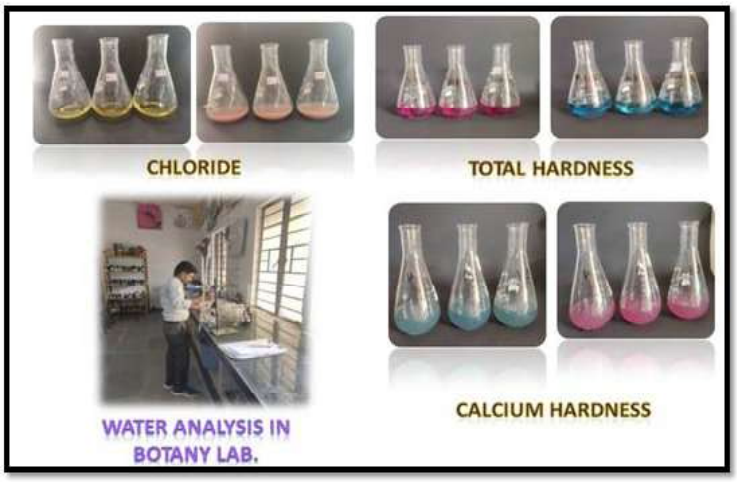

Magnesium Hardness
mg/l
140
120
100
80
60
40
20
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda

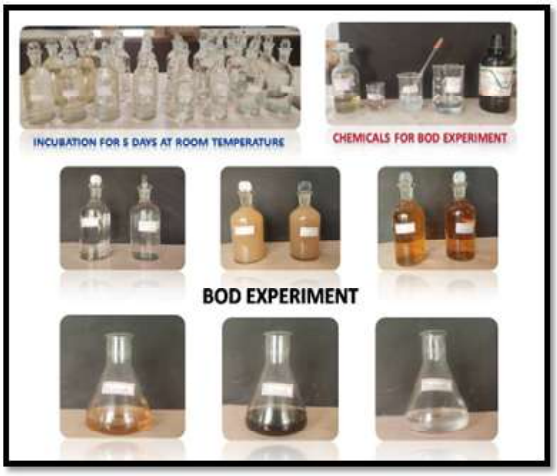

Free CO_2
mg/l
35
30
25
20
15
10
5
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda

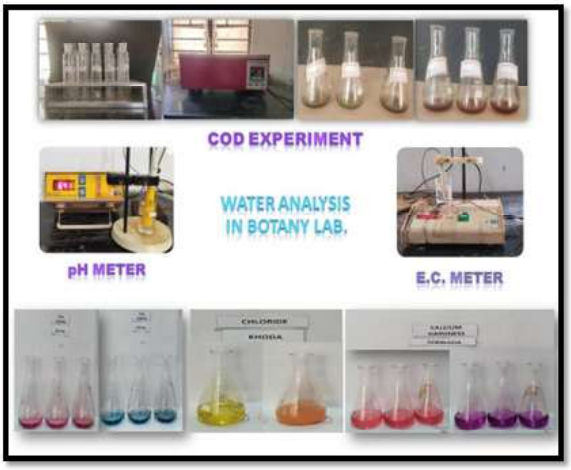

BOD
mg/l
1.5
1
0.5
0
Zalmor
Khoda
Rajpur
Varsada
Mandla
Terwada
Gothda













Total Dissolved Solids (TDS): Total dissolved Solids (TDS) value ranged from 138 to 909 mg/l in winter season, at highest TDS values 909 mg/l Terwada Lake and lowest values 138 mg/l Gothda Lake, Average TDS values 363 mg/l for selected lakes in winter season. TDS value ranged from 193 to 989 mg/l in summer season, at highest TDS values 989 mg/l Terwada Lake and lowest values 193 mg/l Gothda Lake, Average TDS values 641.57 mg/l for selected lakes in summer season. Total dissolved solids describe the number of inorganic salts of calcium, magnesium, sodium etc. and small proportion of organic matter present in the water, where a high value of the same have been reported to be related to acute myocardial infarction as well as ischemic heart diseases in few studies.[10] In this study, TDS values showed a considerable variability ranging from < 10 mg/l - >1500 mg/l.

Total Hardness: Total hardness value ranged from 32 to 167.33 mg/l in winter season, at highest TH values 167.33 mg/l Mandla-1 Lake and lowest values 32 mg/l Gothda Lake, Average TH values 66.29 mg/l for selected lakes in winter season. Total hardness value ranged from 74.67 to 706.67 mg/l in summer season, at highest TH values 706.67 mg/l Rajpur Lake and lowest values 74.67 mg/l Khoda Lake, Average TH values 275.81 mg/l for selected lakes in summer season.

Hardness of water is an important consideration in determining the suitability of water for domestic and industrial uses. Hardness is caused by multivalent metallic cations and with certain anions present in the water to form scale. The principal hardness-causing cations are the divalent calcium, magnesium, strontium, ferrous iron and manganous ions. Hardness was below the permissible limit in all samples and might have caused increased concentration of salts by excessive evaporation as also observed by Bhatt *et al*. (1999)[11].

Calcium Hardness: Calcium ions are important components of plant tissues and participate in various cellular functions. It is also required as a nutrient for various metabolic processes and assists in proper translocation of carbohydrates that facilitates the availability of other ions.[12] During the present study period Calcium hardness varied from 25.33-48.00 mg/l in winter season. The lowest Calcium hardness 25.33 mg/l was observed at Gothda Lake while maximum 48.00 mg/l at Mandla-1, Average calcium Hardness values 32.09 mg/l for selected lakes in winter season and Calcium hardness varied from 17.33-208 mg/l in summer season. The lowest Calcium hardness 17.33 mg/l was observed at Khoda Lake while maximum 208 mg/l at Rajpur Lake, Average calcium Hardness values 105.71 mg/l for selected lakes in summer season.

Magnesium Hardness: Magnesium hardness carryout for study period Magnesium hardness varied from 1.62-29.00 mg/l in winter season. The lowest Magnesium hardness 1.62 mg/l was observed at Gothda Lake while maximum 29.00 mg/l at Mandla-1 Lake in winter season, Average Magnesium Hardness values 8.31 mg/l for selected lakes in winter season and Magnesium hardness varied from 13.93-121.18 mg/l in summer season. The lowest Magnesium hardness 13.93 mg/l was observed at Gothda Lake while maximum 121.18 mg/l at Rajpur Lake in summer season, Average Magnesium Hardness values 41.31 mg/l for selected lakes in summer season.

Chlorides: It occurs naturally in all types of waters. High concentration of chlorides is considered to be the indicators of pollution due to organic wastes of animal or industrial origin. Chlorides are troublesome in irrigation water and also harmful to aquatic life.[13] The chloride content showed very narrow changes in sampling points between seven lakes. The recorded chloride values varied from 27.22-868.57 mg/l in winter season. The lowest chloride values 27.22 mg/l was observed at Zalmor Lake while maximum 868.57 mg/l at Mandla-1 Lake, Average Chloride values 187.31 mg/l for selected lakes in winter season and chloride values varied from 196.43-1367.93 mg/l in summer season. The lowest chloride values 196.43 mg/l was observed at Varsada lake while maximum 1367.93 mg/l at Rajpur lake, Average Chloride values 547.71 mg/l for selected lakes in summer season. Higher concentration of chloride is hazardous to human consumption and creates health problems. Desirable limit of chloride by ISI (1991)[14] for drinking purpose is 250 mg/l.

Free Carbon Dioxide: The amount of free CO_2 in water is generally maintained by diffusion from atmosphere, respiration by animals along with plants and bacterial decomposition of organic matter.[15] During the present study free CO_2 fluctuated between 6.67-23.33 mg/l in winter season., highest being recorded 23.33 mg/l at Gothda Lake and lowest 6.67 mg/l at zalmor and Varsada Lake, Average 12.86 mg/l for selected lakes in winter season and free CO_2 fluctuated between 6.67-31.67 mg/l in summer season, highest being recorded 31.67 mg/l at Gothda Lake and lowest 6.67 mg/l at Varsada Lake, Average 16.19 mg/l for selected lakes in summer season.

Dissolved Oxygen (DO): The dissolved oxygen content is one of the most important factors in stream health. Its deficiency directly affects the ecosystem of a river due to bioaccumulation and biomagnifications. The oxygen content in water samples depends on a number of physical, chemical, biological and microbiological processes. DO values also show lateral, spatial and seasonal changes depending on industrial, human and thermal activity? In the present study, the value of DO range from 0.57-1.97 mg/l in winter season, highest being recorded 1.97 mg/l at Mandla-1 lake and lowest 0.57 mg/l at Rajpur lake, Average 1.20 mg/l for selected lakes in winter season and the value of DO range from 1.13-2.23 mg/l in summer season, highest being recorded 2.23 mg/l at Mandla-1 lake

and lowest 1.13 mg/l at Rajpur Lake, Average 1.56 mg/l for selected lakes in summer season.

Biological Oxygen Demand (BOD): is a measure of the oxygen in the water that is required by the aerobic organisms. The biodegradation of organic materials exerts oxygen tension in the water and increases the biochemical oxygen demand.[16] BOD is the amount of oxygen required by the living organisms engaged in the utilization and ultimate destruction or stabilization of organic water.[17] The value for BOD was found to 0.23-1.20 mg/l in winter season, highest being recorded 1.20 mg/l at Mandla-1 Lake and lowest 0.23 mg/l at Gothda Lake, Average 0.75 mg/l for selected lakes in winter season and the value of BOD range from 0.57-1.50 mg/l in summer season, highest being recorded 1.50 mg/l at Mandla-1 Lake and lowest 0.57 mg/l at Gothda Lake, Average 1.00 mg/l for selected lakes in summer season. Higher values of BOD and lower values of DO indicate more amount of organic matter present in sewage.[18]

Chemical Oxygen Demand (COD): is a measure of the oxidation of reduced chemicals in water. It is commonly used to indirectly measure the amount of organic compounds in water. The measure of COD determines the quantities of organic matter found in water. This makes COD useful as an indicator of organic pollution in surface water.[19,20] The chemical oxygen demand (COD) marked difference among varies site. COD range observed value from 32-288 mg/l in winter season, highest being recorded 288 mg/l at Terwada Lake and lowest 32 mg/l at Khoda and Rajpur Lake, Average 118.86 mg/l for selected lakes in winter season and COD range observed value from 32-224 mg/l in summer season, highest being recorded 224 mg/l at Zalmor and Khoda Lake and lowest 32 mg/l at Mandla-1 lake, Average 132.57 mg/l for selected lakes in summer season.

CONCLUSION

The present study reveals that the water quality of Kankrej taluka lake is deteriorated. It was due to directly mixing of the domestic sewage and human activity in lakes. To improve the quality of water, sewage treatment plants are essential. Therefore, the discharged of effluents before treatment and other waste into the lakes should be controlled and enforced.

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